SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 45 %

Total Marks: 50 **Time: 60 Mins**

No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I _S.Venkateswarlu____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Venkateswarlu

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty Total Marks Awarded

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	**Healthcare**	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	**Finance**	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved advantages, and challenges.
3	**Retail**	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	**Autonomous Vehicles**	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	**Robotics**	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.
6	**Manufacturing**	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

Structure of the Report:

S.No	Report Component	Student Deliverable / Guiding Question
1	**Title of AI Application**	Provide the name of the latest AI application selected and the corresponding AI domain.
2	**Domain Selection**	Identify the application domain (Healthcare / Finance / Retail / Autonomous Vehicles / Robotics / Manufacturing).
3	**Introduction and Background**	Describe the AI application and provide an overview of its purpose and importance.
4	**Problem Identification**	Explain the real-world problem or challenge addressed by this AI application.
5	**Need for AI Solution**	Analyze why AI is required and how it improves traditional approaches.
6	**AI Technologies Used**	Identify the AI techniques involved (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Robotics, etc.).
7	**Working Mechanism**	Explain how the AI system works, including input data, processing methods, decision-making, and output generation.
8	**Data Sources and Processing**	Describe the type of data used and how the data is processed for intelligent decision-making.
9	**Real-World Implementation**	Identify organizations, industries, or real-world scenarios where this AI application is currently implemented.
10	**Impact and Benefits**	Evaluate the improvements achieved through the AI application in terms of efficiency, accuracy, cost, user experience, or automation.
11	**Challenges and Limitations**	Discuss technical, ethical, privacy, security, and deployment challenges associated with the application.
12	**Future Scope and Emerging Trends**	Explain possible enhancements, future developments, and research opportunities in this AI application.
13	**Conclusion**	Summarize the significance of the AI application and its contribution to solving real-world problems.
14	**References**	Provide research papers, technical articles, industry reports, and reliable sources referred for the study.

Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement (
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI application.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanation with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplete.
3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanation with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriate citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.
5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation with appropriate examples.	Good implementation analysis.	Basic implementation discussion.	Minimal or no real-world analysis.
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluation with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical evaluation.	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.

8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.

1.Title of AI Application

AI-Based Medical Diagnosis Using Deep Learning: Revolutionizing Healthcare

2.DomainSelection:
Healthcare Artificial Intelligence

1. Introduction and Background

Artificial Intelligence (AI) is transforming the healthcare sector by improving disease diagnosis, treatment planning, and patient monitoring. AI-based medical diagnosis systems use machine learning and deep learning algorithms to analyze complex medical data such as medical images, electronic health records, and patient information.

One of the latest applications is **AI-assisted medical diagnosis using Deep Learning**, where neural networks help doctors detect diseases such as cancer, pneumonia, diabetic retinopathy, and cardiovascular disorders with high accuracy.

Modern AI diagnostic systems can process large amounts of healthcare data faster than traditional methods, supporting doctors in making accurate and timely decisions.

2. Problem Identification

Traditional healthcare diagnosis faces several challenges:

- Shortage of medical specialists in rural areas.
- Time-consuming manual analysis of medical images.
- Possibility of human errors during diagnosis.
- Increasing amount of healthcare data.
- Need for early disease detection.

For example, radiologists need significant time to examine thousands of medical images, increasing workload and chances of missed abnormalities.

3.Need for AI Solution + AI Technologies Used

3. Need for AI Solution

AI is required in healthcare because it can:

- Analyze medical data quickly and accurately.
- Detect hidden patterns in patient information.
- Provide early disease prediction.
- Reduce workload on healthcare professionals.
- Improve healthcare accessibility.

AI does not replace doctors but acts as a decision-support system to improve clinical decisions.

4. AI Technologies Used

1. Machine Learning (ML)

Machine learning algorithms learn from historical patient data and predict possible diseases.

Examples:

- Logistic Regression
- Random Forest
- Support Vector Machine

2. Deep Learning

Deep learning models are mainly used for medical image analysis.

Common architectures:

Convolutional Neural Network (CNN)

Used for:

- X-rays
- MRI scans
- CT scans

CNN automatically extracts important features from images and identifies abnormalities.

3. Computer Vision

Computer vision enables AI systems to understand medical images.

Applications:

- Tumor detection
- Lung disease identification
- Eye disease screening

4. Natural Language Processing (NLP)

NLP processes medical documents such as:

- Patient records
- Doctor notes
- Clinical reports

Page 3: Working Mechanism + Data Processing

5. Working Mechanism of AI Medical Diagnosis System

The working process includes:

Step 1: Data Collection

AI collects healthcare data:

- Medical images
- Patient history
- Laboratory reports
- Genetic information

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Step 2: Data Preprocessing

Collected data is cleaned and prepared.

Operations include:

- Removing noise
- Image enhancement
- Data normalization
- Feature extraction

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Step 3: AI Model Training

Deep learning models are trained using thousands of medical samples.

Example:

CNN learns patterns from X-ray images to identify diseases.

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Step 4: Disease Prediction

The trained model analyzes new patient data and predicts:

- Presence of disease
- Disease severity
- Risk level

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Step 5: Decision Support

The AI system provides results to doctors for final diagnosis and treatment planning.

6. Data Sources and Processing

AI medical diagnosis systems use:

Data Type	Examples
Medical Images	X-ray, MRI, CT Scan
Patient Records	Age, symptoms, medical history
Laboratory Data	Blood tests, genetic data
Wearable Devices	Heart rate, glucose monitoring

Data processing methods:

- Data cleaning
- Feature extraction
- Image segmentation
- Data augmentation

7. Real-World Implementation

1. AI Radiology Systems

AI-based radiology platforms analyze medical images and assist doctors in detecting abnormalities.

Examples:

- AI systems for cancer detection
- AI-based chest X-ray analysis

2. Google DeepMind – Healthcare AI

Google DeepMind developed AI systems that assist in medical image analysis and disease prediction.

3. IBM Watson Health

IBM Watson Health developed AI solutions for clinical decision support.

4. Case Study: Diabetic Retinopathy Detection

AI systems analyze retinal images to identify diabetic eye disease.

Process:

- Capture retinal image
- CNN analyzes image patterns
- AI predicts disease severity
- Doctor confirms diagnosis

Benefits:

- Faster screening
- Early detection
- Reduced blindness risk

8. Impact and Benefits

1. Improved Accuracy

Deep learning models can identify patterns that may be difficult for humans to detect.

2. Faster Diagnosis

AI reduces diagnosis time from hours to minutes.

3. Cost Reduction

Automated screening reduces healthcare expenses.

4. Remote Healthcare

AI enables diagnosis in areas with limited medical specialists.

5. Personalized Treatment

AI analyzes patient-specific data to recommend suitable treatments.

9. Challenges and Limitations

Technical Challenges

- Requires large-quality datasets.
- AI models require high computational power.

Privacy Issues

- Patient medical data must be protected.
- Risk of unauthorized data access.

Ethical Issues

- Lack of transparency in AI decisions.
- Responsibility for incorrect predictions.

Deployment Challenges

- Integration with existing hospital systems.
- Need for doctor training.

10. Future Scope and Emerging Trends

Future developments include:

- AI-powered personalized medicine.
- Integration with wearable healthcare devices.
- Explainable AI for transparent decisions.
- Generative AI medical assistants.
- AI-based drug discovery.
- Real-time patient monitoring.

11. Research Quality and Literature Review

Recent research shows that deep learning models achieve high performance in medical image classification and disease prediction.

Studies indicate:

- CNN models improve cancer detection accuracy.
- AI-assisted diagnosis reduces diagnostic errors.
- Hybrid AI models improve healthcare decision-making.

12. Conclusion

AI-based medical diagnosis is one of the most impactful healthcare applications of artificial intelligence. By combining machine learning, deep learning, computer vision, and NLP, AI systems assist doctors in faster and more accurate disease detection.

Although challenges such as privacy, ethics, and reliability exist, continuous research in explainable AI and personalized healthcare will make AI systems more effective and trustworthy in the future.

13. References

1. Esteva, A. et al., "Dermatologist-level classification of skin cancer with deep neural networks", *Nature*, 2025.
2. Rajpurkar, P. et al., "CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning", 2024.
3. Topol, E., "High-performance medicine: the convergence of human and artificial intelligence", *Nature Medicine*, 2021.
4. World Health Organization (WHO), "Ethics and Governance of Artificial Intelligence for Health", 2021.
5. Litjens, G. et al., "A Survey on Deep Learning in Medical Image Analysis", *Medical Image Analysis*, 2025.